

Experiment 1: Probability Theory

Title

Probability Calculation using the Breast Cancer Wisconsin Diagnostic Dataset

Objective

To calculate the prior probability of each class (Malignant and Benign) and use these probabilities to predict the class of a new data instance.

Dataset Description

The Breast Cancer Wisconsin Diagnostic Dataset contains medical measurements collected from breast cancer patients. The dataset has two target classes:

- Malignant (Cancerous)
- Benign (Non-Cancerous)

The dataset consists of 569 samples.

Step 1: Load the Dataset

The dataset was loaded using the Scikit-learn library. The target column was extracted to identify the class labels.

Step 2: Count the Class Instances

The number of Malignant and Benign samples was counted.

- Malignant = 212
- Benign = 357

Step 3: Calculate Prior Probabilities

The probability of each class was calculated using the formula:

$P(\text{Class}) = \text{Number of instances in class} / \text{Total number of instances}$

The obtained probabilities were:

- $P(\text{Malignant}) = 0.3726$
- $P(\text{Benign}) = 0.6274$

Step 4: Predict the Class

The class having the highest prior probability was selected as the prediction for a new data instance.

Since the probability of the Benign class was greater than the Malignant class, the predicted class was Benign.

Analysis

The dataset contains more Benign samples than Malignant samples. Therefore, the prior probability of Benign is higher. This simple probability-based prediction uses only class frequencies and does not consider feature values.

Conclusion

The prior probabilities of both classes were successfully calculated. Based on these probabilities, the new instance was predicted as Benign.